

Carl McMillan

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Profile

Software engineer and AI researcher with over ten years of experience in software development and DevOps, and a recently submitted PhD in Artificial Intelligence. My research applies deep learning and computer vision to mussel farm monitoring. I currently have four peer-reviewed publications. I enjoy turning AI research into practical tools that are useful in everyday tasks, and I am driven by solving problems in simple and efficient ways.

Skills

Languages

Python, Bash, Java, SQL

Deep Learning

PyTorch, CNNs (ResNet, DenseNet), Vision Transformers (ViT), YOLO object detection

Computer Vision & Data Science

OpenCV, scikit-learn, NumPy, pandas, Matplotlib

Cloud, HPC & DevOps

AWS, HPC clusters (Slurm, GPU), Terraform, Docker, Kubernetes, GitLab CI, Jenkins, Git, Linux

Work Experience

Doctoral Researcher at Victoria University of Wellington

Jul 2023 - Sep 2026

- Developed deep learning models (CNNs, Vision Transformers) for predicting the buoyancy of mussel farm floats from video, including BuoyancyNet, published in the Journal of the Royal Society of New Zealand.
- Built an end-to-end pipeline from raw mussel farm video to float buoyancy alerts, combining object detection and buoyancy prediction models.
- Assessed the feasibility of deploying the alerting system on a solar-powered monitoring buoy.
- Ran large-scale training and evaluation experiments on a Slurm-managed GPU cluster.

Research Assistant at Victoria University of Wellington

Apr 2023 - Jun 2023

- Annotated a mussel float detection and buoyancy dataset used by multiple researchers for the Centre for Data Science and Artificial Intelligence aquaculture research group at VUW.
- Trained an object detection model on this dataset and produced a video demonstration showcasing the Centre's capabilities.

DevOps Engineer at Fronde

Sep 2020 - Sep 2022

- Built AWS infrastructure for integrated microservice environments using Terraform.
- Built GitLab CI/CD pipelines for a monorepo microservices project.
- Set up monitoring and alerting using AWS CloudWatch, Prometheus, Grafana and Slack integrations.

Technical Consultant (Development/DevOps) at Westpac

Jun 2017 - Sep 2020

- Built Java microservices in Kubernetes, using Kafka for distributed messaging, to integrate proprietary bank card processing software running on older platforms.
- Implemented encrypted storage of financial transaction data in databases and Kafka topics for PCI DSS compliance.
- Automated server builds and installation of proprietary bank card processing software using Chef and Jenkins.
- Set up monitoring and alerting for these systems using Elastic Stack, Grafana and PagerDuty.
- Provided on-call production support for these systems.

Software Developer at Datacom

Jun 2011 - Jun 2017

- Developed web-based data visualisation software for tracking the availability of Chorus resources.
- Developed integration components for the Chorus fibre ordering system as part of the development team.
- Developed AMS integrations for remote operations on smart meters.
- Provided on-call production support for the team's existing applications.
- Improved the team's development processes through automation and ChatOps.

Software Developer at Wialus

Nov 2009 - Feb 2010

- Summer of Code internship.
- Created C++ programs for ETL processes for mobile phone signal data.

- Created a C++ program for calculating elevations along any great circle path.

Biosecurity Officer at GWRC

Jul 2006 - Nov 2007

- Controlled pest animals in the Wellington region and helped the public with pest animal problems.
- Set up poison and trapping programs for volunteers to take over.
- Carried out mapping with GPS devices and ArcGIS software.
- Maintained the database and generated reports for a large conservation project in the Wainui water catchment.
- Conducted trials for possum control methods.

Education

Victoria University of Wellington

Submitted Sep 2026

- PhD in Artificial Intelligence
- Thesis: Deep Learning for Building a Buoyancy Alerting System in Aotearoa New Zealand Mussel Farms

Victoria University of Wellington

Completed Oct 2023

- Postgraduate Diploma in Science in Artificial Intelligence

Victoria University of Wellington

Completed Dec 2011

- Conjoint BSc/BCA in Computer Science/Mathematics/Money & Finance

Victoria University of Wellington

Completed May 2004

- BSc in Ecology & Biodiversity/Marine Biology

Publications

- Carl McMillan, Junhong Zhao, Bing Xue, Ross Vennell, and Mengjie Zhang. Local Context-Aware Buoyancy Prediction for Mussel Farm Floats. *PRICAI 2025: Trends in Artificial Intelligence*, Wellington, New Zealand, November 2025, LNCS vol. 16452, pp. 459-474. DOI: 10.1007/978-981-95-7072-0_31
- Carl McMillan, Junhong Zhao, Bing Xue, Ross Vennell, and Mengjie Zhang. BuoyancyNet: A Deep Learning Approach for Assessing Float Buoyancy in Mussel Aquaculture. *Journal of the Royal Society of New Zealand*, vol. 55, no. 6, 2025, pp. 2013-2041. DOI: 10.1080/03036758.2025.2488415
- Carl McMillan, Junhong Zhao, Bing Xue, Ross Vennell, and Mengjie Zhang. Deep Learning-Based Buoyancy Prediction for Intelligent Mussel Farm Monitoring. *2024 39th International Conference on Image and Vision Computing New Zealand (IVCNZ)*, Christchurch, New Zealand, December 2024, pp. 1-6. DOI: 10.1109/IVCNZ64857.2024.10794479
- Carl McMillan, Junhong Zhao, Bing Xue, Ross Vennell, and Mengjie Zhang. Improving Buoy Detection with Deep Transfer Learning for Mussel Farm Automation. *2023 38th International Conference on Image and Vision Computing New Zealand (IVCNZ)*, Palmerston North, New Zealand, November 2023, pp. 1-6. DOI: 10.1109/IVCNZ61134.2023.10343657